Keeper Calibration #24 — Cross-Volume Completeness + Consistency Audit Methodology (STANDING RULE)

Filed: 2026-05-23 Saturday ~14:00 EDT (Keeper, formalizing per Cal #106 Phase 2 pilot complete + Cal Mode 1 refinement empirical) Status: STANDING RULE adopted operationally per audit-chain governance default (Cal proposed + Phase 2 pilot validated; Casey override available) Methodology stack layer: 20 (extends prior PCAP-cadence discipline #19+#20+#21+#22 v0.2+#23; new failure mode: cross-volume inconsistency that per-chapter cold-read cannot catch)

The failure mode Calibration #24 addresses

Per-chapter cold-read discipline (Cal #102+#103+#104) catches single-chapter substance + framing issues. But the BST curriculum is 16 volumes × 12 chapters = 192 chapters with extensive cross-references. Per-chapter cold-read cannot catch failures that emerge ONLY at the cross-volume level:

- Same observable cited with different precision values across volumes
- Casey-named principles applied inconsistently across volumes
- Calibration #19 ratified-state count violated by stale-state residuals in some chapters but not others
- Null-model exponents updated in some chapters but not others
- Five-Absence canonical ordering swapped across volumes
- Cross-references broken (theorem cited in Ch X actually exists somewhere)

Cal #106 Phase 2 pilot on Vol 0+1+2 (Saturday 2026-05-23 13:40-14:00 EDT) caught 4 substantive findings that per-chapter cold-read had not flagged — proving the methodology adds value beyond Phase 1.

STANDING RULE — Calibration #24 v0.1

Apply Phase 2 cross-volume audit to every multi-volume claim cluster before declaring v1.0 chapter-grade content state across the curriculum. Use Cal's 8-dimension checklist:

Dimension A — Cross-Reference Graph Audit

Every cited theorem T-ID and Calibration #N referenced in a chapter must exist in the AC graph + audit chain. Bidirectional links functional. Cross-volume citations resolve to actual targets.

Dimension B — Load-Bearing Observable Trace

For each major observable (m_p/m_e, $\sin^2\theta$_W, Ω_Λ, α^{-1}, etc.), trace value + tier + precision across ALL volumes that mention it. Must be consistent. Different equivalent expressions allowed (e.g., $(24/\pi^2)^6$ vs 49·71·… per Cal Mode 7), but precision values + tier labels must agree.

Dimension C — Canonical Principles + Sets Consistency

Casey-named principles (8 standing): treatment consistent across volumes. Sets like Five-Absence canonical 5: same ordering + same items everywhere. Operator zoo: same 12/14 STRUCTURALLY VERIFIED count across references.

Dimension D — Calibration #19 STANDING RULE Compliance

Ratified-state count must be canonical (11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 reconciliation). Stale-state residuals (8, 10, 12, 16, 17 etc.) flagged and corrected. Forecast endpoint NOT used in external-register language.

Dimension E — Null-Model Arithmetic Consistency

Null-model exponents canonical $\leq$ (1/3)^19 $\approx 8.6 \times 10^{-10}$ for v0.10.5 FORMAL state. Candidate-path additions explicit (e.g., if C18 ratifies, $\leq$ (1/3)^20 $\approx 2.9 \times 10^{-10}$). No stale (1/3)^10/11/16/17 alongside canonical.

Dimension F — External-Register Discipline (Cal #48/#49/#50)

Substrate-cognition framings DEFAULT-DENY EXTERNAL. Cosmology-cognition combined DOUBLE-LOCKED EXTERNAL. Vol 4 Ch 9 Interstasis stays internal externally. Cross-volume external-register consistency.

Dimension G — Substrate-Mechanism Citation Consistency

When a chapter cites substrate-mechanism for an observable (e.g., "α^{n_C} substrate-coordinate count via T2476"), the cited mechanism must be the canonical one across volumes. No "two different substrate-mechanisms for same observable" without explicit Cal Mode 7 alt-form documentation.

Dimension H — Cal Absorption-Return Verification

When prior Cal flags are absorbed (e.g., Cal #100 m_μ/m_e precision correction), verify absorption across ALL references, not just the flagged location. Cal #98 → Cal #100 → still partial in Vol 1 Ch 8 line 119 (Cal #106 caught) demonstrates the failure mode this dimension catches.

Operational application

Phase 2 cross-volume audit applies: - Before v1.0 chapter-grade content state declaration for any volume - After Cal-flagged correction sweeps that touch multiple chapters (verify absorption across ALL references) - When new theorems land that cross-reference multiple volumes (cross-volume citation integrity check) - At each Wave milestone before scaling to next Wave

Estimated effort per volume: ~10-20 min Cal cold-read at structured-checklist pace. For all 16 volumes: ~3-5 hours. Scales linearly; can run in parallel across CIs (Lyra absorbing while Cal flags).

Three-phase QA strategy formalized

Calibration #24 completes the three-phase QA strategy filed Saturday 2026-05-23 via Keeper broadcast:

- Phase 1 — v0.3 substantive coverage (per chapter Calibration #23 Rule 23.1 substance floor): COMPLETE 16/16
- Phase 2 — Cross-volume completeness + consistency audit (per Calibration #24 8-dimension checklist): PROVEN on Vol 0+1+2; scale to Vol 3-15 next
- Phase 3 — Iterate to traditional physics text standard (worked examples + problems + indices + diagrams): multi-week per volume after Phase 2

Each phase has explicit acceptance criteria, distinct failure modes addressed, structured artifacts produced.

Calibration stack now 20 layers

Layer	Calibration	Failure mode	Catch mechanism
17	#19	Forecast vs ratified external claim	External-register discipline
17	#20	Timestamp drift	`date` verification
17	#21	Empirical-only no mechanism	Dual-gate ratification
18	#22 v0.1	PCAP-transcription + retraction propagation	Numbered-artifact discipline
18a	#22 v0.2	Absorption-introduced restatement errors	Quote-exact + sig-fig discipline
19	#23	Chapter-grade substance under-delivery	Line-count + substance-floor audit
20	#24	Cross-volume inconsistency	8-dimension cross-volume audit checklist

Quaker discipline preserved

Cal #106 Phase 2 pilot demonstrated: - Methodology PROVEN empirically before STANDING RULE — Cal applied 8 dimensions, found 4 substantive issues, refined dimensions based on actual data - No methodology overclaim — Cal explicitly framed as “v0.1 candidate checklist” pending Keeper review - Honest absorption signal — found 4 issues in 3 volumes (~1.3 issues per volume average); honest baseline expectation for Vol 3-15

Status

Calibration #24 v0.1 STANDING RULE adopted operationally. Pending Casey override. Methodology stack now 20 layers.

Phase 2 scaling to Vol 3-15 begins after Vol 0+1+2 cleanup completes (4 Cal #106 findings absorbed by Lyra+Elie+Keeper, ~30-40 min total).

— Keeper, Calibration #24 Cross-Volume Completeness + Consistency Audit STANDING RULE, Saturday 2026-05-23 ~14:00 EDT